

Implementation of the High-Precision Tanh Construction

Purpose. This note records the actual construction used in the paper. Once the bandwidth and the column scales are fixed, the implementation is simply one rank-truncated least-squares solve. The analytic construction is used only to (i) choose a useful hidden scale and (ii) determine reference coefficient envelopes for column scaling; the analytic reference coefficients themselves are never computed.

1. Geometry and bandwidth selection

For an interior resolution N , set

$$h = \frac{2}{N}, \quad R = \lceil a_H \sqrt{N} \rceil, \quad x_j = -1 + jh, \quad j = -R, \dots, N + R.$$

There are $W = N + 2R + 1$ hidden units. The network is

$$q_c(x) = b + \sum_{j=-R}^{N+R} w_j \tanh(\gamma(x - x_j)), \quad c = (b, w_{-R}, \dots, w_{N+R}).$$

The relative bandwidth is $\lambda = \gamma h$. Select λ before fitting. From the sampled target, estimate the amplitude-weighted mean absolute angular frequency

$$\hat{\omega} = \frac{\sum_{k \in \mathcal{K}_M} |\hat{b}_k| |\omega_k|}{\sum_{k \in \mathcal{K}_M} |\hat{b}_k|}, \quad \omega_k = \pi k,$$

using the M distinct period-two samples; omit the repeated right endpoint. The zero mode remains in the denominator and contributes zero to the numerator. If the frequency is known analytically, use it directly.

Given the paper's effective aliasing budget ε_{eff} , choose the high-precision branch of

$$\boxed{2 \exp\left(-\frac{\pi^2}{\lambda_{\text{pred}}}\right) \sinh\left(\frac{2\pi\hat{\omega}}{N\lambda_{\text{pred}}}\right) = \varepsilon_{\text{eff}}} \quad (1)$$

and take the largest admissible solution on the prescribed bracket. This is a scalar root solve, not a network-fit sweep. Finally set

$$\lambda = \lambda_{\text{pred}}, \quad \gamma = \frac{\lambda}{h} = \frac{N\lambda}{2}.$$

Smaller λ suppresses aliasing; once aliasing is within budget, larger λ is preferable because the readout and boundary-correction bounds deteriorate as λ decreases.

2. Reference coefficient envelopes and column scaling

The representation proof constructs an accurate reference readout c^* , but the implementation does *not* evaluate it. We use only deterministic coordinatewise bounds

$$|c_j^*| \leq B\alpha_j,$$

where B bounds the target on its complex neighborhood. They bound how large each coordinate may need to be in an accurate boundary-corrected representation.

Let

$$d_{\text{pole}} = \frac{\pi h}{2\lambda}, \quad D_\lambda = \frac{\pi}{2\lambda} + \frac{4 \log 2}{\pi}, \quad m = \left\lfloor \frac{R+1}{2} \right\rfloor, \quad \zeta = e^{-2\lambda}.$$

For $p_0 = 1$, $p_k = \prod_{\ell=1}^k (1 - \zeta^\ell)$, define

$$L_{i,m} = \frac{\zeta^{[i(i+1)-1]/2}}{p_{i-1}p_{m-i}} \prod_{\substack{1 \leq j \leq m \\ j \neq i}} (1 + \zeta^{j-1/2}), \quad i = 1, \dots, m.$$

There are two contributions.

1. **Baseline quasi-interpolant.** The analytic density satisfies $|a_\gamma(x_j)| \leq B/(\delta - d_{\text{pole}})$, while the baseline tanh weight is $w_j^{\text{base}} = ha_\gamma(x_j)/2$. Hence every hidden coordinate starts with

$$\boxed{\alpha_j^{\text{base}} = \frac{h}{2(\delta - d_{\text{pole}})}}.$$

2. **Boundary correction.** The i th boundary moment is bounded by BhD_λ/δ , and $L_{i,m}$ bounds the absolute amplification from those moments to the corresponding rational-correction coefficient. Converting that coefficient to a tanh weight contributes another factor $1/2$. Therefore, for the m selected outer-halo slots on each side,

$$\alpha_{j_i^\pm} = \alpha_j^{\text{base}} + \frac{hD_\lambda L_{i,m}}{2\delta},$$

while all other hidden slots keep $\alpha_j = \alpha_j^{\text{base}}$. Anchoring the network gives

$$\alpha_b = 1 + \sum_{j=-R}^{N+R} \alpha_j$$

for the output-bias coordinate.

The envelopes are therefore cheap functions of the geometry and require no complex target evaluations. Their practical purpose is **reference scaling**: choose

$$d_j = \sqrt{\alpha_j} \quad (\text{rounded upward in the certified implementation}), \quad D = \text{diag}(d_j).$$

If $a^\star = D^{-1}c^\star$, then

$$\|a^\star\|_2^2 = \sum_j \frac{|c_j^\star|^2}{\alpha_j} \leq B^2 \sum_j \alpha_j, \quad \|Da\|_1 \leq \left(\sum_j \alpha_j\right)^{1/2} \|a\|_2.$$

Thus the analytic witness is bounded in the Euclidean coordinates used by the SVD, and scaled-coordinate control transfers to the native readout. This is the purpose of the envelope calculation.

3. Matrix assembly and one least-squares solve

Use only in-domain observations

$$z_i = -1 + \frac{ih}{s}, \quad i = 0, \dots, sN, \quad M_X = sN + 1, \quad y_i = f(z_i).$$

Form the design matrix $A \in \mathbb{R}^{M_X \times (W+1)}$ with an explicit bias column,

$$A_{i,b} = 1, \quad A_{i,j} = \tanh(\gamma(z_i - x_j)).$$

The halo neurons are ordinary columns of this matrix; no exterior labels are used. Scale the columns and normalize the objective:

$$F = \frac{AD}{\sqrt{M_X}}, \quad y_X = \frac{y}{\sqrt{M_X}}.$$

Then perform one rank-truncated solve

$$a_{\text{fit}} = F_\tau^\dagger y_X, \quad c_{\text{fit}} = Da_{\text{fit}}.$$

The theorem chooses the cutoff so unresolved singular directions cost no more than the current representation/recovery budget; implementation is a standard truncated-SVD least-squares call. The returned network is

$$\hat{q}_W(x) = c_b + \sum_{j=-R}^{N+R} c_j \tanh(\gamma(x - x_j)).$$

In short: estimate a target frequency and choose λ by (1); build the uniform core+halo tanh dictionary; compute the analytic reference envelopes and use them only to scale the columns; then call the truncated least-squares solver once and unscale the result. The longer numerical proof just certifies that this simple implementation inherits the reference approximation up to the stated sampling and floating-point allowance.